

WEEK 4 – ASSIGNMENT

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Title:

Azure Cloud Fundamentals and Data Pipeline Implementation using ADF

Objective:

To understand Azure cloud concepts and build a complete data pipeline using Storage Account and Azure Data Factory.

Assignment Tasks

Task 1:

- Explore Azure Portal
- Create a Resource Group

Deliverable:

- Screenshot of Resource Group

Home > Resource Manager | Resource groups

Create a resource group ...

Basics Tags Review + create

Resource group - A container that holds related resources for an Azure solution. The resource group can include all the resources for the solution, or only those resources that you want to manage as a group. You decide how you want to allocate resources to resource groups based on what makes the most sense for your organization. [Learn more](#)

Subscription * ⓘ	Azure for Students
Resource group name * ⓘ	cei-week4-Himanshu
Region * ⓘ	(Asia Pacific) Central India

Previous

Next

Review + create

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Microsoft Azure

Search resources, services, and docs (G+)

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Home > Resource Manager

Resource Manager | Resource groups

Summarize my costs by service

Export resource groups using Bicep or Terraform

How to manage changes with deployment tools?

Search

Create

Manage view

Refresh

Export to CSV

Open query

Assign tags

Add to service group

Group by none

Resource Manager

All resources

Favorite resources

Recent resources

Resource groups

Tags

Organization

Tools

Deployments

Help

Filter for any field...

Subscription equals all

Location equals all

Add filter

☐	Name ↑	Subscription	Location
☐	cei-week4-Himanshu	Azure for Students	Central India

Showing 1 - 1 of 1. Display count: auto

Give feedback

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Task 2: Storage Setup

Do:

- Create a Storage Account
- Create a Blob Container
- Upload a CSV file

Deliverable:

- Screenshot of container with uploaded file

The screenshot displays the Microsoft Azure portal interface. At the top, the navigation bar shows 'Microsoft Azure' with a search bar and user information. The breadcrumb trail indicates the path: Home > ceihbstorage01_1783836962294 | Overview. The main heading is 'ceihbstorage01' with a 'Storage account' subtitle. A left-hand sidebar lists various management tools like Activity log, Tags, and Storage browser. The main content area is divided into sections: 'Essentials' with metadata (Resource group, Location, Subscription, etc.), 'Properties' with a 'Blob service' section listing features like Hierarchical namespace and Soft delete, and 'Security' and 'Networking' sections with their respective settings. A 'JSON View' link is visible in the top right corner of the main content area.

Microsoft Azure

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AB0BDD0E-5790-4EBA-ASAB-78...

Home > ceihbstorage01_1783836962294 | Overview

ceihbstorage01
Storage account

Search

Upload Open in Explorer Delete Move Refresh Open in mobile CLI / PS Feedback

Overview

Activity log

Tags

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Security + networking

Data management

Settings

Monitoring

Automation

Add or remove favorites by pressing Ctrl+Shift+F

Essentials

Resource group (move) : cei-week4-Himanshu

Location : centralindia

Subscription (move) : Azure for Students

Subscription ID : 12cfae79-a21e-4b1c-aaf6-2a55178e73a5

Disk state : Available

Tags (edit) : Add tags

Performance : Standard

Replication : Locally redundant storage (LRS)

Account kind : StorageV2 (general purpose v2)

Provisioning state : Succeeded

Created : 7/12/2026, 11:48:11 AM

JSON View

Properties Monitoring Capabilities (7) Recommendations (0) Tutorials Tools + SDKs

Blob service

Hierarchical namespace : Disabled

Default access tier : Hot

Blob anonymous access : Disabled

Blob soft delete : Enabled (7 days)

Container soft delete : Enabled (7 days)

Versioning : Disabled

Change feed : Disabled

NFS v3 : Disabled

Allow cross-tenant replication : Disabled

Security

Require secure transfer for REST API operations : Enabled

Storage account key access : Enabled

Minimum TLS version : Version 1.2

Infrastructure encryption : Disabled

Networking

Public network access : Enabled

Public network access scope : Enable from all networks

Private endpoint connections : 0

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A6D8DDDE-5798-4EBA-ASAB-78...

Home > ceihbstorage01_1783836962294 | Overview > ceihbstorage01

ceihbstorage01 Containers ☆ ...

Storage account

Search

Diagnose and solve problems

Access Control (IAM)

Data migration

Events

Storage browser

Storage Mover

Partner solutions

Resource visualizer

Data storage

Containers

Classic file shares

Queues

Tables

> Security + networking

> Data management

> Settings

> Monitoring

+ Add container ↑ Upload ↻ Refresh 🗑️ Delete 🔒 Change access level ↺ Restore containers 📊 Edit columns

Search containers by prefix

Only show active containers

Showing all 3 items

☐	Name	Last modified	Anonymous access level	Lease state	Storage connector
☐	📁 \$logs	7/12/2026, 11:48:34 AM	Private	Available	- ...
☐	📁 processed-data	7/12/2026, 11:51:47 AM	Private	Available	- ...
☐	📁 source-data	7/12/2026, 11:51:35 AM	Private	Available	- ...

Microsoft Azure

Search resources, services, and docs (G+/)

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A6D8DDDE-5798-4EBA-ASAB-78...

Home > ceihbstorage01_1783836962294 | Overview > ceihbstorage01 | Containers

source-data ...

Container

Search

+ Add Directory ↑ Upload 🔒 Change access level ↻ Refresh 🗑️ Delete 📄 Copy 📄 Paste 🔄 Rename ⚙️ Acquire lease ⚙️ Break lease 📊 Edit columns

source-data

Authentication method: Access key (Switch to Microsoft Entra user account)

Add filter

Search blobs by prefix (case-sensitive)

Only show active blobs

Showing all 1 items

☐	Name	Last modified	Access tier	Blob type	Size	Lease state
☐	📄 Sample - Superstore.csv	7/12/2026, 11:52:42 AM	Hot (Inferred)	Block blob	2.18 MiB	Available ...

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Task 3: ADF Basics

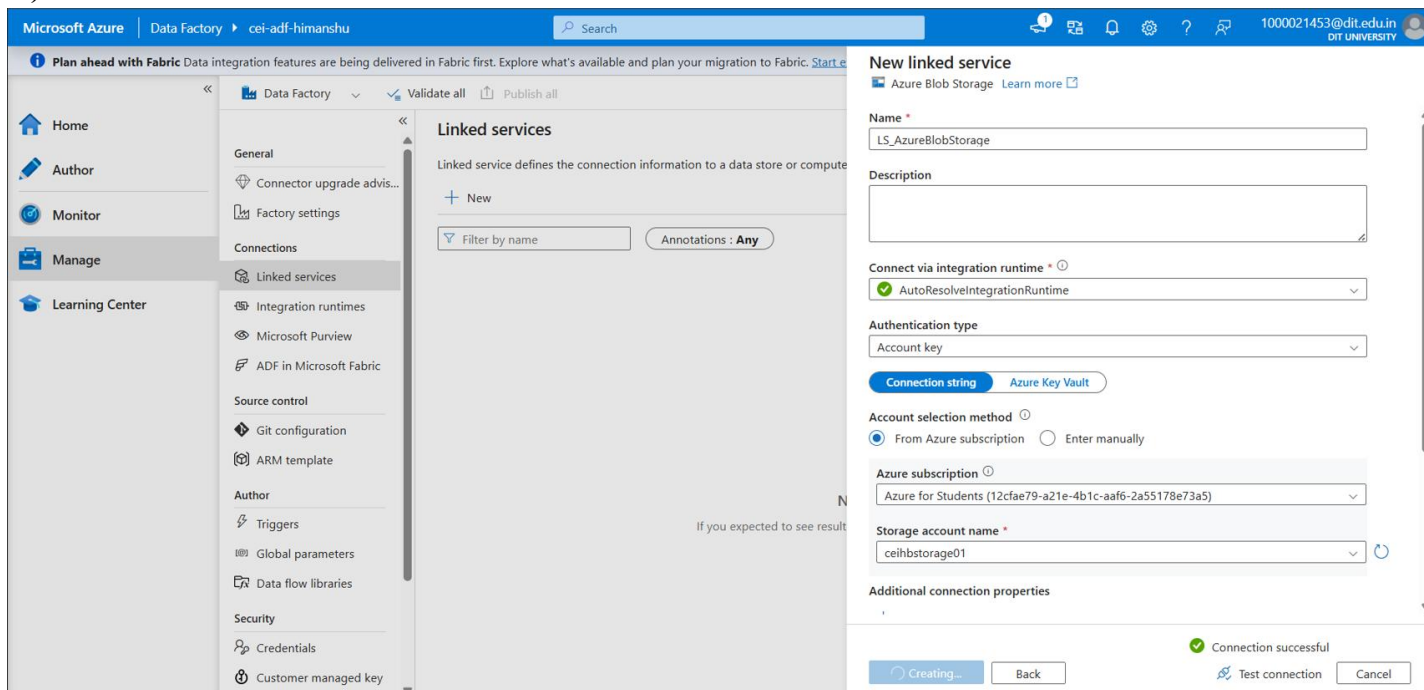
Do:

- Create Azure Data Factory
- Explore ADF UI (Author, Monitor, Manage)
- Create Linked Service (Blob Storage)
- Create datasets (source + destination)
- Use Get Metadata activity

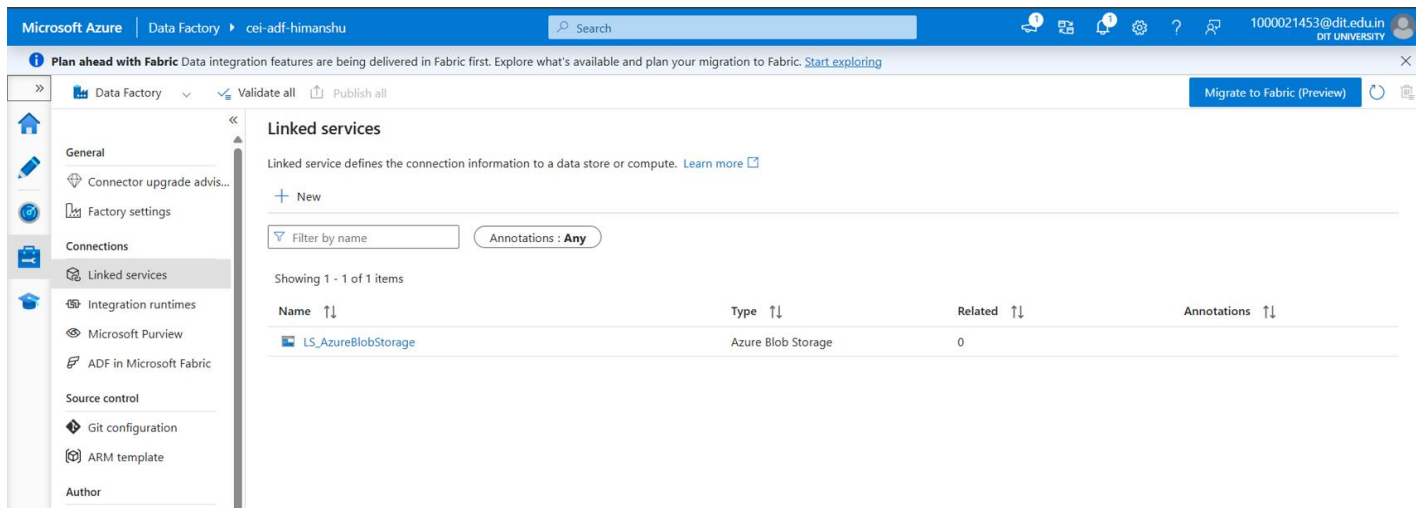
Deliverable:

- Screenshot of Linked Service
- Screenshot of dataset
- Screenshot of Get Metadata activity

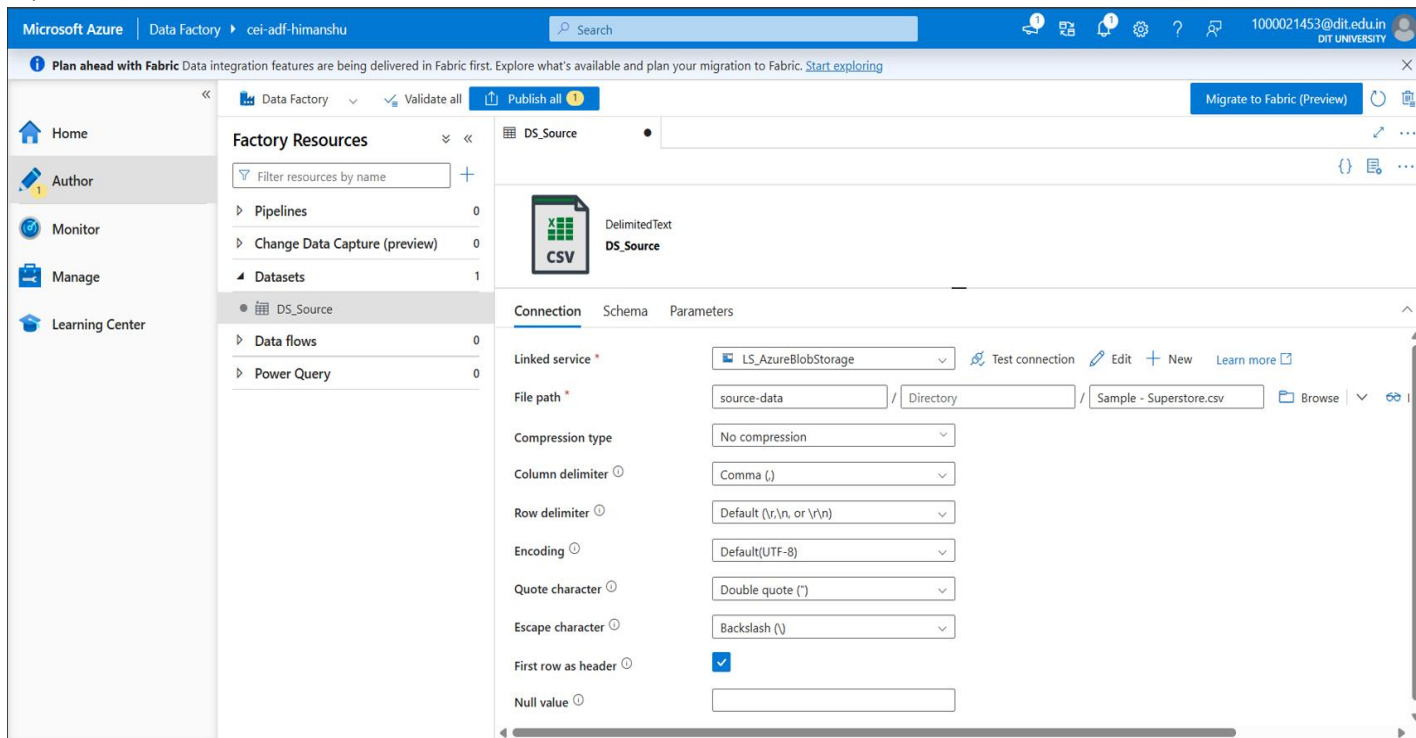
a.) Screenshot of Linked Service



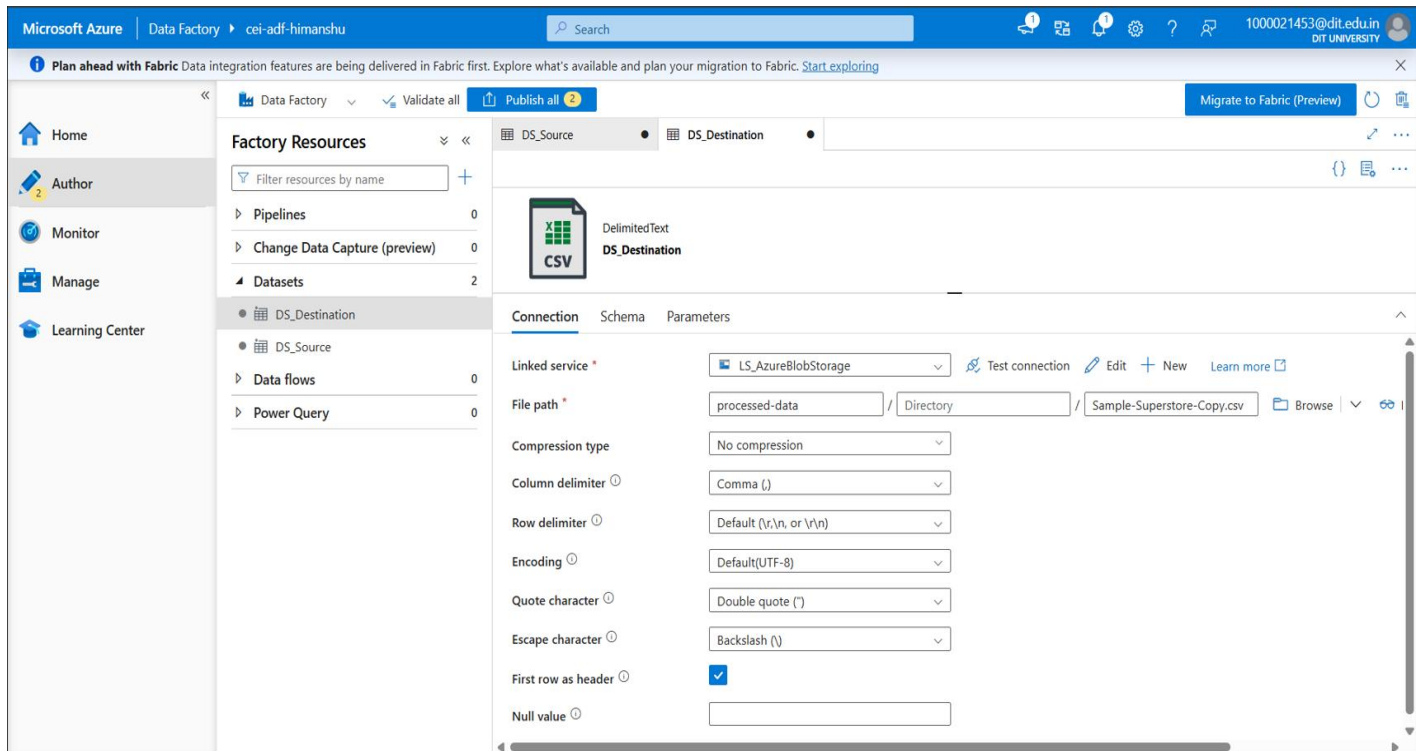
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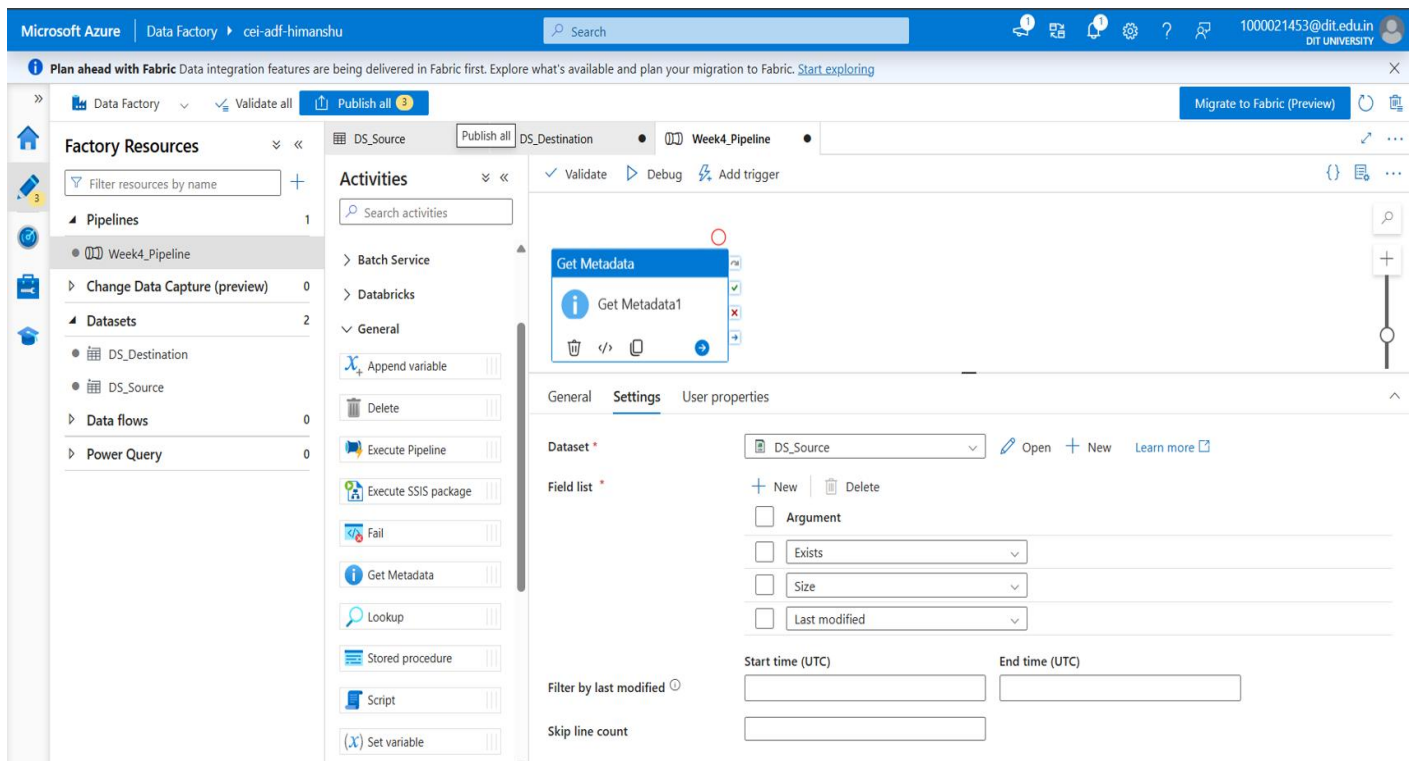
b.) Screenshot of DataSet



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c.) Screenshot of Get Metadata Activity



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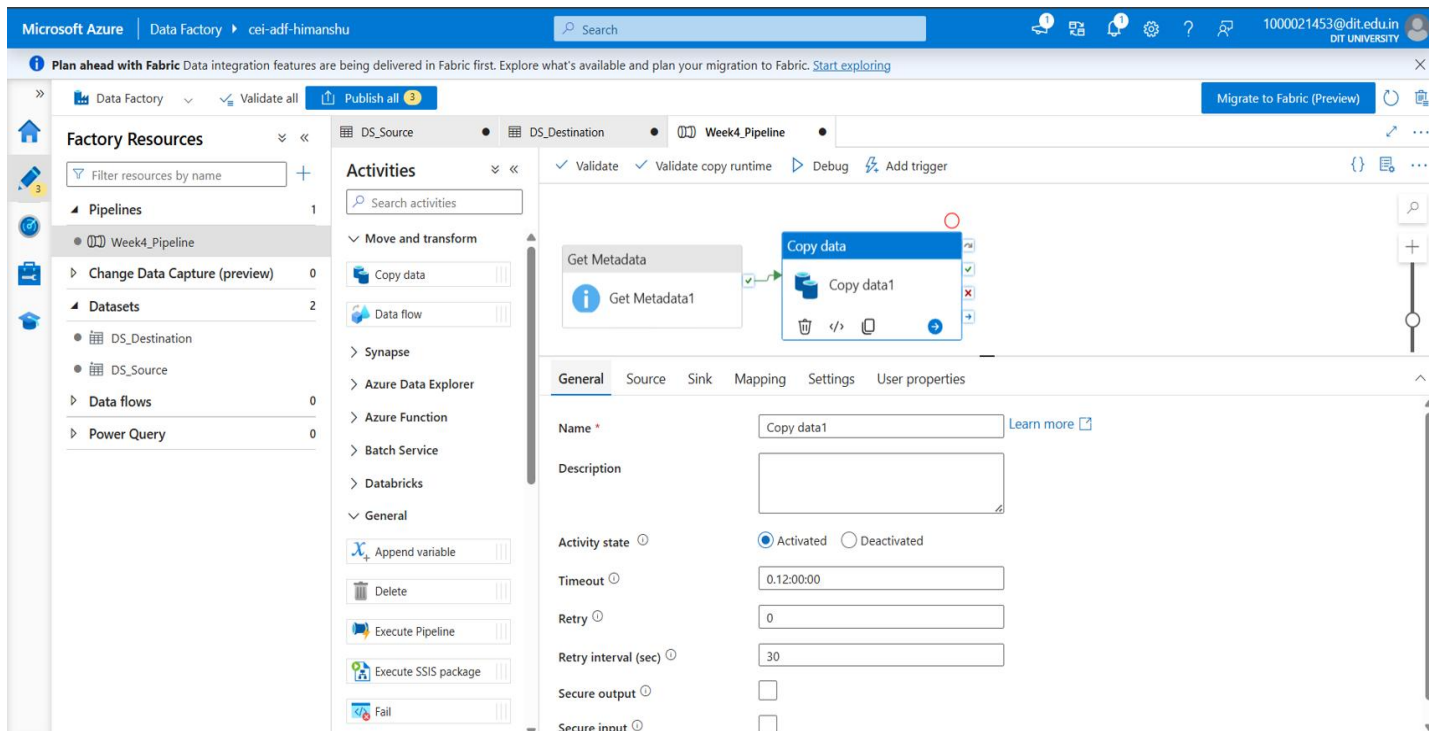
Task 4: Pipeline Development

Do:

- Create pipeline using Copy Data activity
- Configure source and destination
- Add ForEach activity (optional if multiple files)

Deliverable:

- Screenshot of pipeline design



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This screenshot shows the Microsoft Azure Data Factory interface. The top navigation bar includes the Microsoft Azure logo, the path 'Data Factory > cei-adf-himanshu', a search bar, and a user profile with the email '1000021453@dit.edu.in'. A banner at the top promotes Fabric Data integration features. The left sidebar displays 'Factory Resources' with a search bar and a list of items: Pipelines (1), Datasets (2), Data flows (0), and Power Query (0). The 'Week4_Pipeline' is selected. The main canvas shows a pipeline diagram with two activities: 'Get Metadata' (Get Metadata1) and 'Copy data' (Copy data1). The 'Copy data' activity is selected, and its properties are shown in the right-hand pane. The 'Source' tab is active, displaying the following settings: Source dataset is 'DS_Source'; File path type is 'File path in dataset'; Start time (UTC) and End time (UTC) are empty; Filter by last modified is empty; Recursively is checked; Enable partitions discovery is unchecked; Max concurrent connections is empty; Skip line count is empty; and Additional columns is '+ New'.

This screenshot shows the same Microsoft Azure Data Factory interface as the first image, but with the 'Sink' tab selected for the 'Copy data' activity. The 'Sink' tab displays the following settings: Sink dataset is 'DS_Destination'; Copy behavior is 'Preserve hierarchy'; Max concurrent connections is empty; Block size (MB) is empty; Metadata is '+ New'; Quote all text is checked; File extension is '.txt'; and Max rows per file is empty. The pipeline diagram and left sidebar remain the same as in the first image.

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Pipeline Validation

The screenshot displays the Microsoft Azure Data Factory console. The top navigation bar shows 'Data Factory' and 'cei-adf-himanshu'. The left sidebar lists 'Factory Resources' including Pipelines (Week4_Pipeline), Change Data Capture (preview), Datasets (DS_Destination, DS_Source), Data flows, and Power Query. The main area shows the 'Week4_Pipeline' with two activities: 'Get Metadata' and 'Copy data'. The 'Copy data' activity is selected, and its 'General' tab is active, showing fields for Name, Description, Activity state (Activated), Timeout (0.12:00:00), and Retry (0). On the right, a 'Factory validation output' panel shows a green checkmark and the message: 'Your factory has been validated. No errors were found.' A 'Close' button is at the bottom right of this panel.

Pipeline Published

This screenshot shows the same Microsoft Azure Data Factory console as the previous one, but with a 'Publishing completed' notification. The notification is a white box with a green checkmark and the text: 'Publishing completed Successfully published'. The 'Factory validation output' panel is still visible on the right, showing the same validation message. The 'Copy data' activity is still selected in the main area.

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Task 5: Pipeline Execution

Do:

- Run pipeline using Debug/Trigger

Deliverable:

- Screenshot showing pipeline execution (Succeeded)

The screenshot displays the Microsoft Azure Data Factory console. The left sidebar shows the 'Factory Resources' tree with 'Pipelines' expanded, listing 'Week4_Pipeline'. The main area shows the 'Activities' tab for 'Week4_Pipeline', displaying a flow from 'Get Metadata1' to 'Copy data1'. Below the flow, the 'Output' tab shows the 'Pipeline run' details for ID 'f36306f3-1b71-4047-9558-f187ff-12be48', indicating a 'Succeeded' status. A table lists the activities:

Activity name	Activity status	Activity name	Run start	Duration	Integration runtime
Copy data1	Succeeded	Copy data	7/12/2026, 12:47:13 PM	16s	AutoResolveIntegrationRuntime (Central
Get Metadata1	Succeeded	Get Metadata	7/12/2026, 12:46:58 PM	14s	AutoResolveIntegrationRuntime (Central

The screenshot displays the 'Pipeline runs' section of the Microsoft Azure Data Factory console. The left sidebar shows the 'Monitor' tab selected. The main area shows the 'Pipeline runs' list for 'Week4_Pipeline'. The list includes columns for Pipeline name, Run start, Run end, Duration, Status, Triggered by, and Run ID. The status is 'Succeeded'.

Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID
Week4_Pipeline	7/12/2026, 12:46:51	7/12/2026, 12:47:31	40s	Succeeded	Manual trigger	f36306f3-1b71-4047...

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Task 6: IAM Roles

Do:

- Assign roles:
 - Reader
 - Contributor
- Provide access to ADF for Storage

Deliverable:

- Screenshot of role assignment

Microsoft Azure

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DIT UNIVERSITY (DIT.EDU.IN)

Home > ceihbstorage01

ceihbstorage01 | Access Control (IAM)

Storage account

Search

+ Add Download role assignments Edit columns Refresh Delete Feedback

Check access **Role assignments** Roles Deny assignments

Number of role assignments for this subscription 5 4000

Search by name or email Type: All Role: All Scope: All scopes Group by: Role

All (4) Job function roles (1) Privileged administrator roles (3)

Name	Type	Role	Scope	Condition
Owner (2)				
Himanshu Batra 1000021453@dit.edu.in	User	Owner	Subscription (Inherited)	None
Himanshu Batra 1000021453@dit.edu.in	User	Owner	Subscription (Inherited)	None
Contributor (1)				
cei-adf-himanshu e749c8c0-a560-430d-8c36-...	Managed identity	Contributor	Resource group (Inherited)	None
Reader (1)				
cei-adf-himanshu e749c8c0-a560-430d-8c36-...	Managed identity	Reader	This resource	None

Showing 1 - 4 of 4 results.

Add or remove favorites by pressing Ctrl+Shift+F

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Mini Project (Final Task)

Objective

Built an end-to-end Azure Data Factory pipeline to copy a CSV file from Azure Blob Storage while validating metadata before processing.

Implementation

- Created an Azure Storage Account and Blob Containers.
- Uploaded the Superstore CSV file to the source container.
- Configured Azure Blob Storage Linked Service.
- Created Source and Destination Datasets.
- Implemented a Get Metadata activity to validate the source file.
- Used Copy Data activity to copy the CSV into the destination container.
- Assigned Reader and Contributor IAM roles to the Azure Data Factory Managed Identity.
- Successfully executed and monitored the pipeline.

Result

- ✓ Source CSV successfully detected.
- ✓ Metadata validation completed successfully.
- ✓ CSV copied to the destination container.
- ✓ Pipeline executed without any errors.
- ✓ End-to-end Azure Data Factory pipeline completed successfully.

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a.) Screenshot of Pipeline Successfully Executed

The screenshot displays the Azure Data Factory Studio interface. On the left, the 'Factory Resources' pane shows a tree view with 'Pipelines' containing 'Week4_Pipeline'. The main canvas shows the 'Week4_Pipeline' with two activities: 'Get Metadata1' and 'Copy data1', both marked with green checkmarks indicating success. Below the canvas, the 'Output' tab shows the 'Pipeline run' details for ID 'f36306f3-1b71-4047-9558-f187ff12be48', with a status of 'Succeeded'. A table below lists the activities and their execution details.

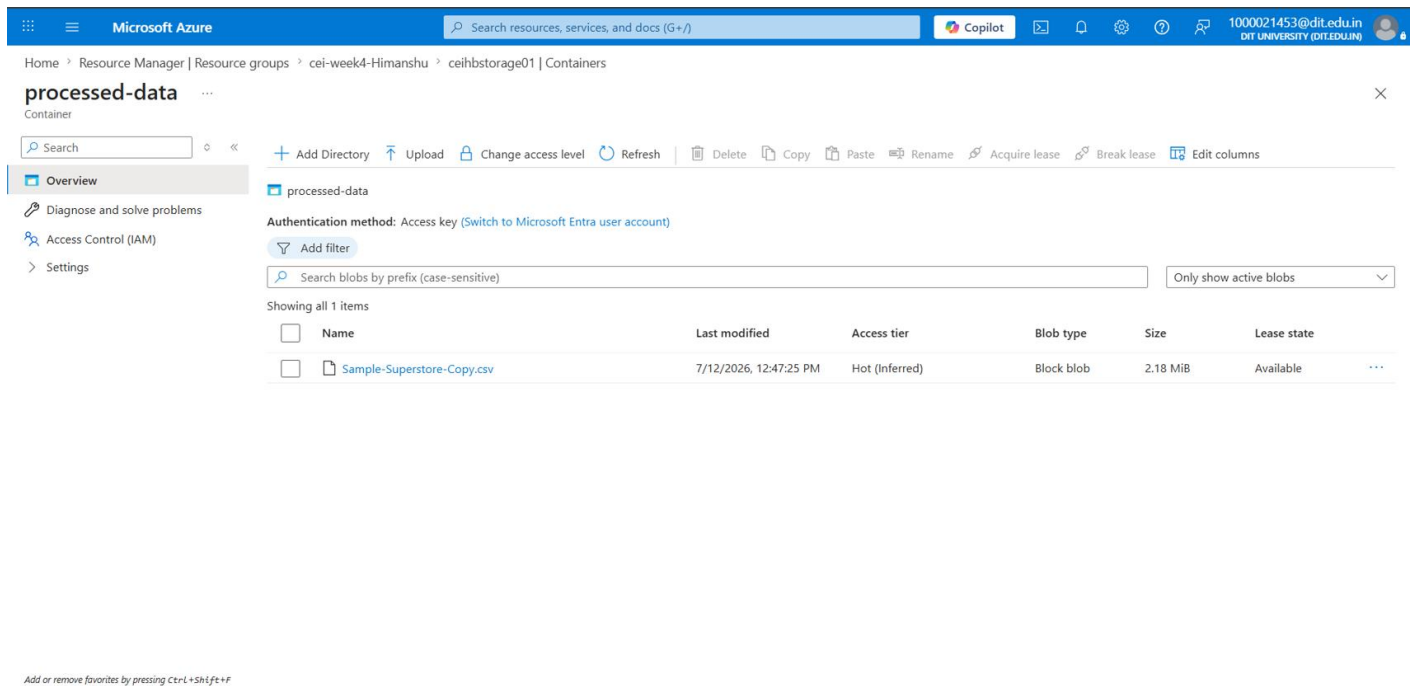
Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/12/2026, 12:47:13 PM	16s	AutoResolveIntegrationRuntime (Central
Get Metadata1	✓ Succeeded	Get Metadata	7/12/2026, 12:46:58 PM	14s	AutoResolveIntegrationRuntime (Central

The screenshot shows the 'Pipeline runs' page in the Azure Data Factory portal. The left navigation pane includes 'Home', 'Author', 'Monitor', 'Manage', and 'Learning Center'. The 'Monitor' section is active, showing 'Pipeline runs'. The main area displays a table of pipeline runs for 'Week4_Pipeline'. The table shows one run that succeeded on 7/12/2026 at 12:47:34, triggered manually. The status is 'Succeeded'.

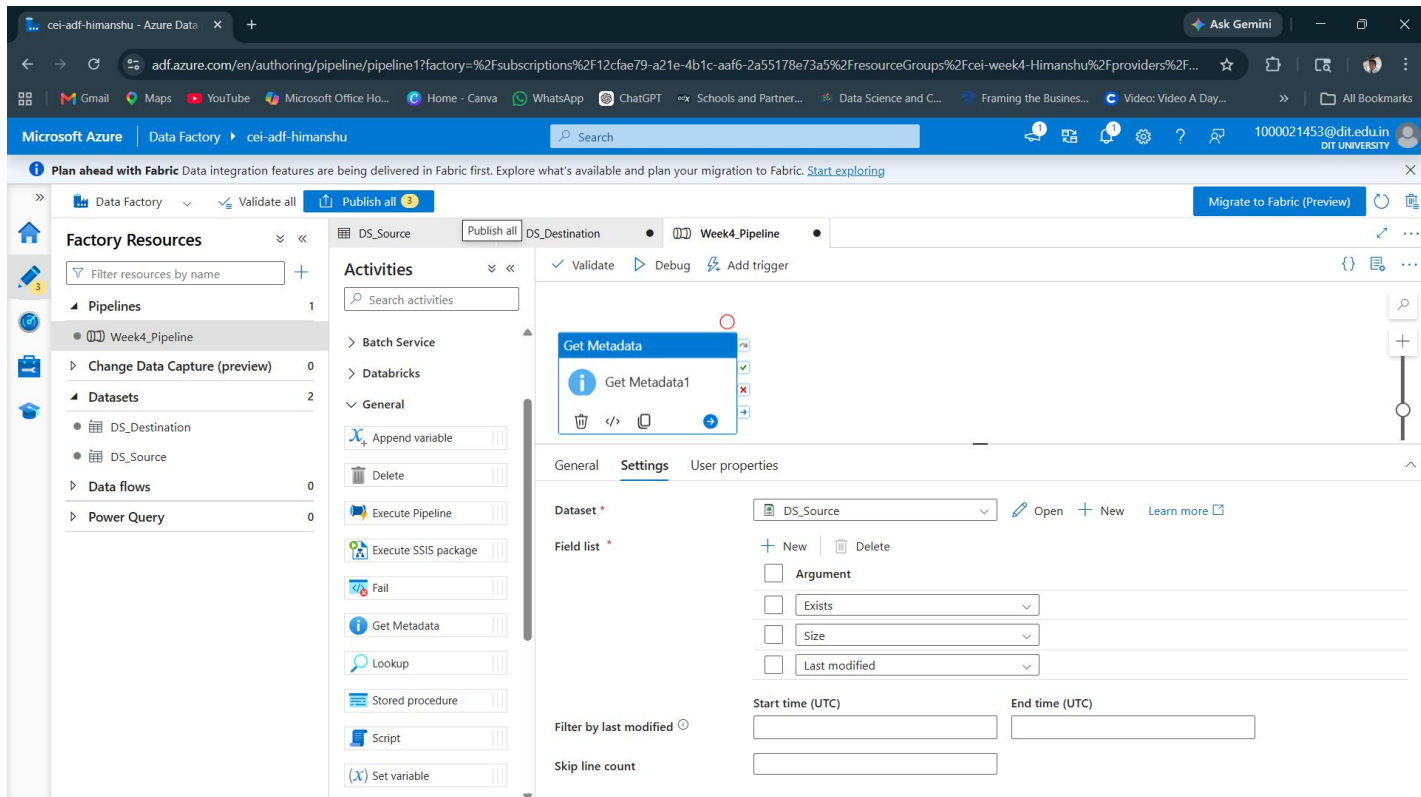
Pipeline name	Run start	Run end	Duration	Status	Triggered by	Run ID	Param
Week4_Pipeline	7/12/2026, 12:46:54	7/12/2026, 12:47:34	40s	✓ Succeeded	Manual trigger	f36306f3-1b71-4047...	

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b.) Screenshot of Data Successfully Copied to Destination



c.) Screenshot of Get Metadata Activity



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d.) Screenshot of Metadata Successfully Validated

The screenshot displays the Azure Data Factory console. On the left, the 'Factory Resources' pane shows the 'Week4_Pipeline' under the 'Pipelines' section. The main canvas shows the pipeline's synapse with two activities: 'Get Metadata1' and 'Copy data1', both marked with green checkmarks indicating success. Below the synapse, the 'Output' tab is active, showing the pipeline run details. The pipeline run ID is 'f36306f3-1b71-4047-9558-f187ff12be48' and its status is 'Succeeded'. A table below lists the activities and their execution details.

Activity name	Activity st...	Activit...	Run start	Duration	Integration runtime
Copy data1	✓ Succeeded	Copy data	7/12/2026, 12:47:13 PM	16s	AutoResolveIntegrationRuntime (Central
Get Metadata1	✓ Succeeded	Get Metadata	7/12/2026, 12:46:58 PM	14s	AutoResolveIntegrationRuntime (Central

Mini Project Summary

An end-to-end Azure Data Factory pipeline was developed to read a CSV file from Azure Blob Storage, validate its metadata using the **Get Metadata** activity, and copy it to a new destination container using the **Copy Data** activity. IAM roles (Reader and Contributor) were configured to enable secure access between Azure Data Factory and Azure Storage. The pipeline executed successfully, and the copied file was verified in the destination (`processed-data`) container, confirming successful data transfer and metadata validation.